ECA1308 – Microwave Engineering

1 A. TE10 Mode Simulation: A WR-90 rectangular waveguide has a = 22.86 mm, b = 10.16 mm and length L = 100 mm. Operate it at f = 10 GHz and excite the dominant TE10 mode in HFSS. Obtain the E-field and H-field distributions, simulated cutoff frequency fc and propagation constant β. Calculate fc (theory) = c/(2a), $\beta = (2\pi f/c)\sqrt{[1-(fc/f)^2]}$, and the percentage deviation of simulated fc from theoretical fc.

B. TE10 versus TM11: Using the same WR-90 waveguide at f = 10 GHz, simulates both TE10 and TM11 modes. Determine fc, β and the E/H-field distributions for each mode. Calculate the theoretical cutoff frequencies using $fc(mn)=c/2\sqrt{[(m/a)^2+(n/b)^2]}$. Compare the results and justify why TE10 propagates at 10 GHz while TM11 does not.

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	E-field distribution	TE10/TM11 field pattern	Record HFSS plot	—
2	H-field distribution	TE10/TM11 field pattern	Record HFSS plot	—
3	Cutoff frequency	TE10 = 6.557 GHz; TM11 ≈ 16.42 GHz	Record HFSS value	% deviation
4	Propagation constant	TE10 ≈ 198.9 rad/m at 10 GHz	Record HFSS value	% deviation
5	Mode decision	TE10 propagates; TM11 is below cutoff	Record observation	—

Aim:

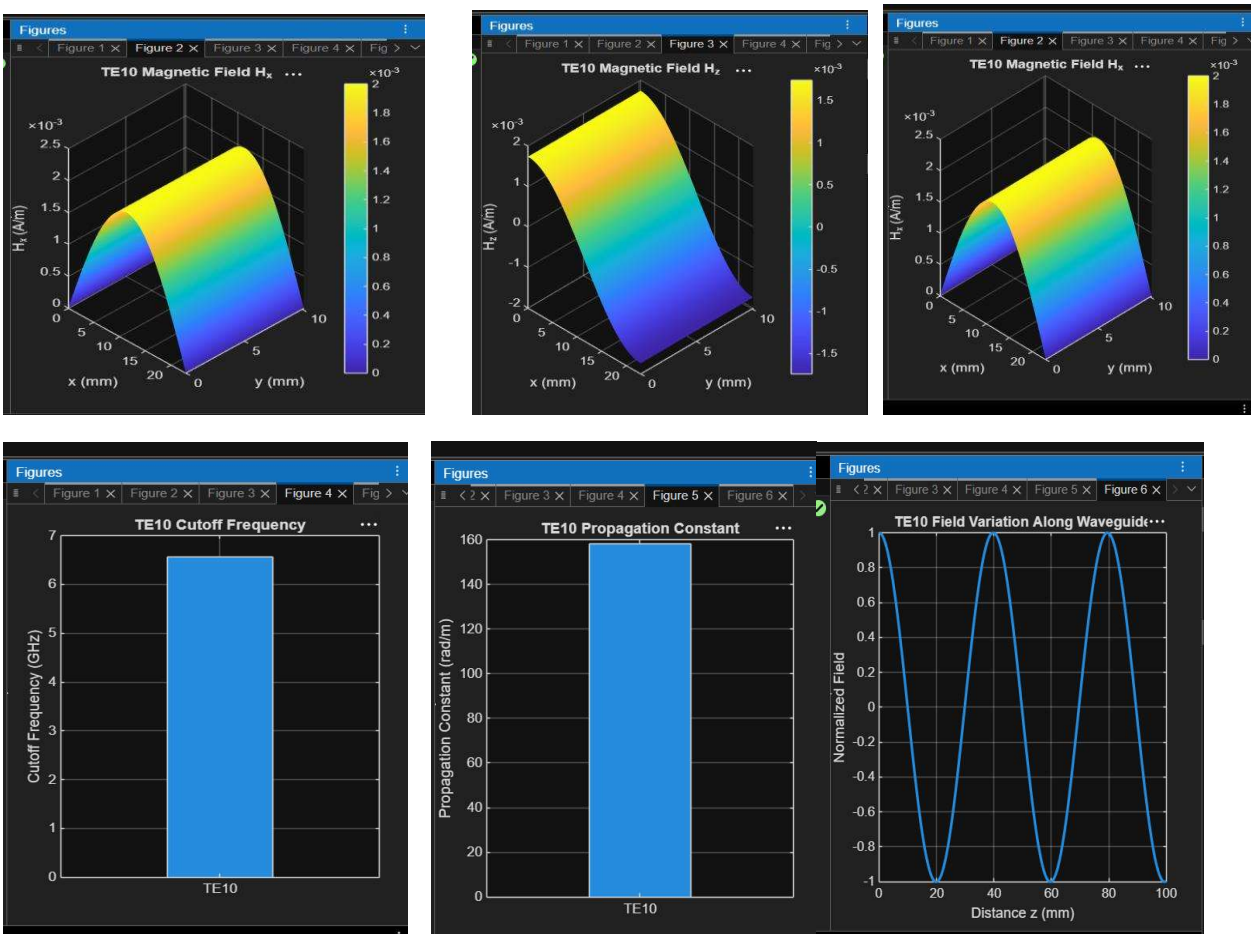
To simulate and compare the TE_{10} and TM_{11} modes in a WR-90 rectangular waveguide operating at 10 GHz using MATLAB, determine their cutoff frequencies and propagation constants, plot their electric and magnetic field distributions, and verify that the TE_{10} mode propagates while the TM_{11} mode is below cutoff and exhibits evanescent decay.

Program: A.

```
clc;
clear;
close all;
% WR-90 waveguide parameters
a = 22.86e-3;
b = 10.16e-3;
L = 100e-3;
f = 10e9;
c = 3e8;
% Theoretical parameters
fc = c/(2*a);
k0 = 2*pi*f/c;
beta = k0*sqrt(1-(fc/f)^2);
fprintf('QUESTION A: TE10 MODE\n');
fprintf('Cutoff Frequency = %.4f GHz\n',fc/1e9);
fprintf('Propagation Constant = %.4f rad/m\n',beta);
% Cross-sectional coordinates
x = linspace(0,a,150);
y = linspace(0,b,100);
[X,Y] = meshgrid(x,y);
% TE10 field patterns
Ey = sin(pi*X/a);
Hx = (beta/(2*pi*f*4*pi*1e-7))*Ey;
Hz = (1/(2*pi*f*4*pi*1e-7)) ...
*(pi/a)*cos(pi*X/a);
% 1. Electric field distribution
figure;
surf(X*1000,Y*1000,Ey);
shading interp;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized E_y');
title('TE10 Electric Field Distribution');
view(45,30);
% 2. Magnetic field Hx distribution
figure;
surf(X*1000,Y*1000,Hx);
shading interp;
colorbar;
```

```
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('H_x (A/m)');
title('TE10 Magnetic Field H_x');
view(45,30);
% 3. Magnetic field Hz distribution
figure;
surf(X*1000,Y*1000,Hz);
shading interp;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('H_z (A/m)');
title('TE10 Magnetic Field H_z');
view(45,30);
% 4. Cutoff frequency graph
figure;
bar(fc/1e9);
set(gca,'XTickLabel',{'TE10'});
ylabel('Cutoff Frequency (GHz)');
title('TE10 Cutoff Frequency');
grid on;
% 5. Propagation constant graph
figure;
bar(beta);
set(gca,'XTickLabel',{'TE10'});
ylabel('Propagation Constant (rad/m)');
title('TE10 Propagation Constant');
grid on;
% 6. Electric field variation along z
z = linspace(0,L,500);
Ez = cos(beta*z);
figure;
plot(z*1000,Ez,'LineWidth',2);
grid on;
xlabel('Distance z (mm)');
ylabel('Normalized Field');
title('TE10 Field Variation Along Waveguide');
```

Output: A



Program: B

clc; clear; close all;

%% WR-90 Rectangular Waveguide Parameters a = 22.86e-3; % Broad dimension (m) b = 10.16e-3; % Narrow dimension (m) L = 100e-3; % Waveguide length (m) f = 10e9; % Operating frequency (Hz)

c = 299792458; % Speed of light (m/s) mu0 = 4*pi*1e-7; eps0 = 1/(mu0*c^2);

w = 2*pi*f; k0 = w/c;

%% 1. Cutoff Frequencies fc_TE10 = (c/2)*sqrt((1/a)^2 + (0/b)^2); fc_TM11 = (c/2)*sqrt((1/a)^2 + (1/b)^2);

```
fprintf('Operating frequency = %.2f GHz\n', f/1e9); fprintf('TE10 cutoff frequency = %.4f GHz\n', fc_TE10/1e9); fprintf('TM11 cutoff frequency = %.4f GHz\n', fc_TM11/1e9);

%% 2. Propagation Constants if f > fc_TE10 beta_TE10 = k0*sqrt(1-(fc_TE10/f)^2); else beta_TE10 = 0; end

if f > fc_TM11 beta_TM11 = k0*sqrt(1-(fc_TM11/f)^2);  alpha_TM11 = 0; else beta_TM11 = 0; kc_TM11 = 2pi*fc_TM11/c; alpha_TM11 = sqrt(kc_TM11^2-k0^2); end

fprintf('TE10 beta = %.4f rad/m\n', beta_TE10); fprintf('TM11 beta = %.4f rad/m\n', beta_TM11); fprintf('TM11 attenuation constant = %.4f Np/m\n', ... alpha_TM11);

if f > fc_TE10 fprintf('TE10: PROPAGATING MODE\n'); else fprintf('TE10: BELOW CUTOFF\n'); end

if f > fc_TM11 fprintf('TM11: PROPAGATING MODE\n'); else fprintf('TM11: BELOW CUTOFF (EVANESCENT)\n'); end

%% 3. Cross-Section Grid x = linspace(0,a,200); y = linspace(0,b,120); [X,Y] = meshgrid(x1e3,y1e3);

% Convert grid back to metres Xm = X1e-3; Ym = Y1e-3;

%% 4. TE10 Electric and Magnetic Field Patterns % Normalized field patterns for the TE10 mode Ey_TE10 = sin(piXm/a); Hz_TE10 = cos(piXm/a);

% Plot normalized field distributions figure('Color','w','Name','TE10 Field Distributions');

subplot(1,2,1); surf(X,Y,Ey_TE10,'EdgeColor','none'); view(2); axis tight; colorbar; xlabel('x (mm)'); ylabel('y (mm)'); title('TE_{10} Electric Field |E_y|'); colormap parula;

subplot(1,2,2); surf(X,Y,Hz_TE10,'EdgeColor','none'); view(2); axis tight; colorbar; xlabel('x (mm)'); ylabel('y (mm)'); title('TE_{10} Longitudinal H_z Pattern'); colormap parula;

sgtitle('WR-90 Waveguide: TE_{10} at 10 GHz');

%% 5. TM11 Electric Field Pattern % TM11 longitudinal electric-field pattern Ez_TM11 = sin(pi*Xm/a).*sin(piYm/b);

figure('Color','w','Name','TM11 Field Distribution'); surf(X,Y,Ez_TM11,'EdgeColor','none'); view(2); axis tight; colorbar; xlabel('x (mm)'); ylabel('y (mm)'); title('TM_{11} Normalized Longitudinal Electric Field'); colormap parula;

%% 6. Field Behaviour Along Waveguide Length z = linspace(0,L,500);
```

% TE10: traveling-wave phase variation TE10_field = cos(beta_TE10*z);

% TM11: evanescent field decay below cutoff TM11_field = exp(-alpha_TM11*z);

figure('Color','w','Name','Propagation Comparison');

plot(z*1e3,TE10_field,'LineWidth',1.8); hold on; plot(z*1e3,TM11_field,'LineWidth',1.8);

grid on; xlabel('Distance z (mm)'); ylabel('Normalized field'); title('TE_{10} and TM_{11} Behaviour at 10 GHz'); legend('TE_{10}: cos(\beta z)', ... 'TM_{11}: exp(-\alpha z)', ... 'Location','best');

%% 7. Cutoff Frequency Comparison figure('Color','w','Name','Cutoff Comparison');

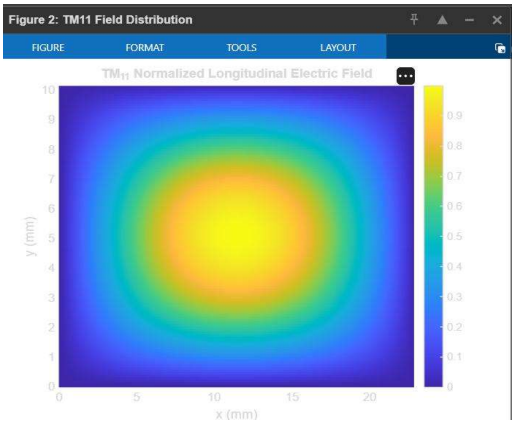
bar([fc_TE10,fc_TM11]/1e9); hold on; yline(f/1e9,'r--','Operating frequency = 10 GHz', ... 'LineWidth',1.5);

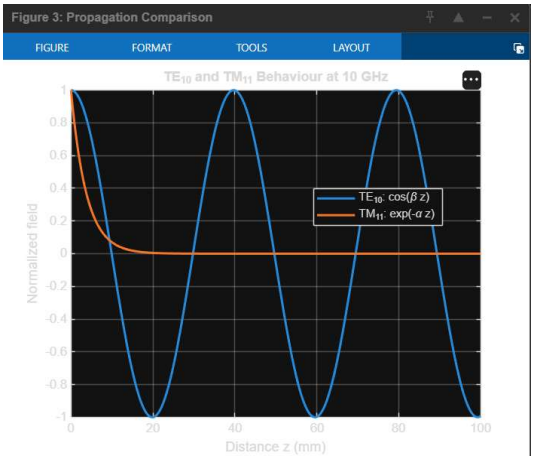
set(gca,'XTickLabel',{'TE_{10}','TM_{11}'}); ylabel('Frequency (GHz)'); title('Cutoff Frequencies vs Operating Frequency'); grid on;

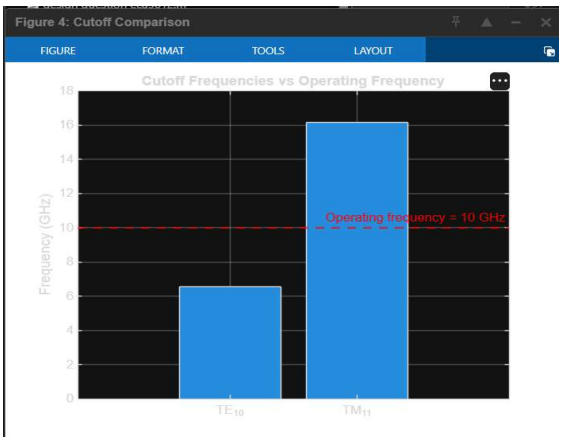
%% 8. Theoretical Summary fprintf('\n--- THEORETICAL SUMMARY ---\n'); fprintf('TE10 cutoff : %.4f GHz\n',fc_TE10/1e9); fprintf('TM11 cutoff : %.4f GHz\n',fc_TM11/1e9); fprintf('TE10 beta : %.4f rad/m\n',beta_TE10); fprintf('TM11 attenuation : %.4f Np/m\n',alpha_TM11); fprintf('Waveguide length : %.2f mm\n',L*1e3);

Output: B









Result:

The cutoff frequencies were found to be approximately 6.56 GHz for TE_{10} and 16.42 GHz for TM_{11}. At 10 GHz, TE_{10} propagates through the WR-90 waveguide, whereas TM_{11} is below cutoff and exhibits evanescent decay.